Name: Bharath Arun Gandhimani **Student ID:** 35501308
Teaching Associate: Michael Niemann & Ayden Zhou, Applied 7

Project Title: Powering the Path: A Spatio-Temporal Analysis of Renewable Energy Infrastructure, Educational Investment, and Household Mobility in Victoria (2024–2026)

Introduction

Victoria aims for 100% renewable energy by 2035. This goal is shifting where people work and changing how they live. To support this, the state is investing in Renewable Energy Zones (REZs) and regional Tech Schools [2]. Solar power already contributes 16% of the state's target [1]. I am researching how these new hubs affect regional travel patterns. I want to find out if these changes help people travel more sustainably or if they make transport inequality worse.

Motivation

I am fascinated by Victoria's transition from coal to renewable energy, as it reflects a major global shift toward green power. [3]. While I am excited that this change creates new regional jobs, I am also concerned about how it might change the way people travel for work. By analyzing travel data from 2024–2025, I want to see if our new infrastructure truly supports sustainable habits or if it accidentally makes transport inequality worse.

Questions

1. How do travel patterns for work and school differ in the areas where Victoria is building new green energy?
2. Is there a relationship between the economic conditions of a region and how people choose to travel to new energy hubs?
3. Does the placement of new schools in a region reflect the daily travel of the workforce?

Data source(s)

- **VISTA 2024-2025 (Relational):** This data represents the daily travel habits of Victorians and includes temporal attributes like start times and categorical details like trip purpose and transport mode. This dataset will be used for all three questions.
  - **Trips** (~150k rows, 40 cols): https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/f8d93c74-cf02-48d7-a368-fc207c0b3c4f/download/trips_vista_2024_2025.csv
  - **Persons** (~82k rows, 51 cols): https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/2a38732d-04dc-4af5-8a52-f281a8ecebb7/download/person_vista_2024_2025.csv
  - **Households** (~32k rows, 30 cols): https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/1b73f806-1565-4243-96d7-78f9002c3e8f/download/household_vista_2024_2025.csv
- **State Budget Map Data 2023-24 (Spatial):** It contains geographical attributes (latitude and longitude) and **quantitative** investment values. This data represents the specific locations and funding amounts for renewable energy projects and regional Tech Schools.This dataset will be used to answer Question 1 and Question 3
  - https://discover.data.vic.gov.au/dataset/ed30724b-6ae7-4713-aff3-4f00b3666684/resource/71ca8b10-5a4e-483d-b3a3-a94421f4ef8b/download/state-budget-2023-24-budget-website-map-data.csv
- **ABS SEIFA 2021 LGA Indexes (Tabular):** It provides ordinal attributes, such as socio-economic deciles and ranks. This data represents the economic status of different local government areas, allowing me to see if energy investments reach disadvantaged communities. This dataset is used to answer Question 2.
  - https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-areas-seifa-australia/2021/Local%20Government%20Area%2C%20Indexes%2C%20SEIFA%202021.xlsx

References

[1] Foley, E. (2026, February 12). *Solar energy chips in 16% of Victoria's renewable energy target*. PV Magazine Australia. https://www.pv-magazine-australia.com/2026/02/12/solar-energy-chips-in-16-of-victoria-renewable-energy-target/ [Accessed on: 15.03.2026 (checked 15.03.2026)].

[2] VicGrid. (2025). *2025 Victorian Transmission Plan*. Department of Energy, Environment and Climate Action. https://engage.vic.gov.au/victransmissionplan [Accessed on: 15.03.2026 (checked 15.03.2026)].

[3] Clean Energy Council. (2025). *Clean Energy Australia Report 2025*. https://cleanenergycouncil.org.au/news-resources/clean-energy-australia-report-2025 [Accessed on: 15.03.2026 (checked 15.03.2026)].